

Nicola Bottura, PhD

WINDOWS LOW-LEVEL C++ ENGINEER · ENDPOINT SECURITY & MALWARE ANALYSIS

Rome, Italy

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Summary

Current RD at Prisma s.r.l., working on malware analysis and conducting scientific research with the Sap4Sec research group at Sapienza University of Rome, where I recently (Jan. 2026) obtained my PhD. 5+ years of experience in malware analysis and reverse engineering, binary instrumentation, and fuzzing techniques. Main author and developer of PFuzzer, a DynamoRIO-based, 17k+ C/C++ LoC coverage-guided analysis framework for environment-sensitive Windows malware—protected by Italian, European, and (pending) US patents—currently being industrialized as a commercial malware analysis platform. Published in relevant security venues (IEEE EuroSP, DIMVA) and collaborated with researchers across multiple institutions. This background has also led me to develop expertise in vulnerability research and offensive security tradecraft.

Work Experience

Prisma s.r.l.

Rome, Italy

RESEARCH & DEVELOPMENT

Nov. 2025 - Present

- Designing and implementing PFuzzer as a sandbox-integrated malware analysis platform for commercial deployment (Windows, C++), focusing on robustness and low-overhead dynamic instrumentation.
- Automating reverse engineering workflows for Windows malware using program analysis techniques (coverage-guided exploration, behavioral telemetry, anti-analysis detection).

CINI Consorzio Interuniversitario Nazionale per l'Informatica

Rome, Italy

RESEARCH FELLOW

March 2022 - May 2022

- Research on novel methods to automate reverse engineering efforts in malware with program analysis.

Education

Sapienza University of Rome

Rome, Italy

PHD IN ENGINEERING IN COMPUTER SCIENCE

May. 2022 - Jan. 2026

- **Thesis title:** Engaging with Malware: Coverage-guided Fuzzing and Dynamic Introspection to Analyze Environment Sensitive and Obfuscated Windows Malware.
- Visited for 3 months University of Twente in Enschede (NL) and collaborated with Prof. Andrea Continella

Sapienza University of Rome

Rome, Italy

M.Sc. IN CYBERSECURITY

Sep. 2019 - Jan. 2022

- **Thesis title:** Revisiting API Interposition Techniques.

Unimore (University of Modena and Reggio Emilia)

Modena, Italy

B.Sc. IN COMPUTER SCIENCE

Sep. 2015 - Jul. 2019

- **Thesis title:** Design and Development of a Mechanism to Reduce the Guaranteed Latency of the BFQ Scheduler.
- Contributed with a [commit] for Linux with my patch to BFQ

Technical Skills

Programming Languages

C, modern C++, Python, x86-64 Assembly, PowerShell, Bash

Windows Internals

Low-level Windows development, Win32 API, Native API, Memory management, COM, Registry

Reverse Engineering

IDA Pro, DBI, Binary Ninja, Ghidra, WinDbg, Static/Dynamic analysis

Malware Analysis

Behavioral analysis, anti-analysis techniques, Obfuscation, Malware PoC development, EDR, anti-tampering

Vulnerability Research

Fuzzing, Harness development, Dynamic Program Analysis, Memory safety

Tooling

DynamoRIO, Intel Pin, AFL, libDFT, YARA, Git

Publications

2025	PDF	Pfuzzer: Practical, Sound, and Effective Multi-path Analysis of Environment-sensitive Malware with Coverage-guided Fuzzing N. Bottura, D. C. D’Elia, L. Querzoni <i>IEEE EuroS&P</i>
2025	PDF	All Right Then, (Don’t) Keep Your Secrets: Exposing API Hashing in Malware N. Bottura, G. Di Pietro, Y. Yamada, D. C. D’Elia, L. Querzoni <i>DIMVA</i>

Patents

2024	PDF	Methods and Systems for Analyzing Environment Sensitive Malware with Coverage-guided Fuzzing N. Bottura; D. C. D’Elia <i>Europe: EP4312401, Italy: IT202200015966A1, Soon US</i>
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Projects

PFuzzer

DynamoRIO-based coverage-guided fuzzer (17k+ LoC in C/C++).
Published; granted a patent; ongoing work toward industrial applicability (currently owned by Prisma s.r.l., code cannot be shared).

Unprotect Project

Contributed with four novel anti-analysis techniques found in malware - link.
Unprotect is an open source project that gathers knowledge on anti-analysis techniques in malware.

Offensive Tradecrafting

Maintain a public repository with PoC prototypes and write-ups supporting analysis and detection tooling - link.
Includes malware techniques found during my research and other techniques I found interesting in re-developing.

Services

2026	Artifact Evaluation Committee Member, Link	<i>NDSS'27</i>
2026	Artifact Evaluation Committee Member, Link	<i>USENIX'26</i>
2026	Artifact Evaluation Committee Member, Link	<i>ACM CCS'26</i>
2026	Reviewer	<i>ACM DTRAP</i>
2025	Artifact Evaluation Committee Member, Link	<i>NDSS'26</i>
2025	Artifact Evaluation Committee Member, Link	<i>ACM CCS'25</i>